

Heterogeneous yet stable $V\delta 2^{(+)}$ T-cell profiles define distinct cytotoxic effector potentials in healthy human individuals

Paul L. Ryan^{a,b}, Nital Sumaria^{a,1}, Christopher J. Holland^{a,1}, Claire M. Bradford^a, Natalia Izotova^a, Capucine L. Grandjean^a, Ali S. Jawad^c, Lesley A. Bergmeier^d, and Daniel J. Pennington^{a,2}

^aCentre for Immunobiology, Blizard Institute, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, London E1 2AT, United Kingdom; ^bCentre for Adult Oral Health, Institute of Dentistry, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, London E1 2AD, United Kingdom; ^cRheumatology Department, Royal London Hospital, London E1 1BB, United Kingdom; and ^dCentre for Clinical and Diagnostic Oral Sciences, Institute of Dentistry, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, London E1 2AT, United Kingdom

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Human $\gamma\delta$ T cells display potent responses to pathogens and malignancies. Of particular interest are those expressing a $\gamma\delta$ T-cell receptor (TCR) incorporating TCR δ -chain variable-region-2 [$V\delta 2^{(+)}$], which are activated by pathogen-derived phosphoantigens (pAgs), or host-derived pAgs that accumulate in transformed cells or in cells exposed to aminobisphosphonates. Once activated, $V\delta 2^{(+)}$ T cells exhibit multiple effector functions that have made them attractive candidates for immunotherapy. Despite this, clinical trials have reported mixed patient responses, highlighting a need for better understanding of $V\delta 2^{(+)}$ T-cell biology. Here, we reveal previously unappreciated functional heterogeneity between the $V\delta 2^{(+)}$ T-cell compartments of 63 healthy individuals. In this cohort, we identify distinct “ $V\delta 2$ profiles” that are stable over time; that do not correlate with age, gender, or history of phosphoantigen activation; and that develop after leaving the thymus. Multiple analyses suggest these $V\delta 2$ profiles consist of variable proportions of two dominant but contrasting $V\delta 2^{(+)}$ T-cell subsets that have divergent transcriptional programs and that display mechanistically distinct cytotoxic potentials. Importantly, an individual’s $V\delta 2$ profile predicts defined effector capacities, demonstrated by contrasting mechanisms and efficiencies of killing of a range of tumor cell lines. In short, these data support patient stratification to identify individuals with $V\delta 2$ profiles that have effector mechanisms compatible with tumor killing and suggest that tailored $V\delta 2$ -profile-specific activation protocols may maximize the chances of future treatment success.

human $\gamma\delta$ T cells | $V\delta 2^{(+)}$ T cells | antitumor cytotoxicity | functional heterogeneity | human immunology

Human $\gamma\delta$ T cells display potent responses to pathogens and malignancies (1–3). Of particular interest are those that express TCR δ -chain variable-region-2 [$V\delta 2^{(+)}$] (4). These cells are uniquely activated by low-molecular-weight nonpeptide phosphoantigens (pAgs), such as microbial-derived (E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HMB-PP) (5, 6). $V\delta 2^{(+)}$ T cells are also activated by isopentenyl pyrophosphate, a phosphoantigen (pAg) that accumulates in transformed eukaryotic cells or cells exposed to aminobisphosphonates such as zoledronate (7). The mechanism of pAg-mediated activation of $V\delta 2^{(+)}$ T cells is still unclear, but likely involves pAg association with butyrophilin 3A1 (BTN3A1) (8, 9).

Subsequent to pAg stimulation, $V\delta 2^{(+)}$ T cells can display significant cytokine-dependent functional plasticity. Indeed, Th1-like, Th2-like, Th17-like, and Treg-like effector characteristics have all been reported (10–14). Moreover, $V\delta 2^{(+)}$ T cells demonstrate efficient cytolytic capacity (15–17) and function as antigen-presenting cells if activated under appropriate conditions (18). These functional potentials underpin the role of $V\delta 2^{(+)}$ T cells in diverse immune scenarios. $V\delta 2^{(+)}$ T cells respond vigorously to

Mycobacterium tuberculosis (19) and *Plasmodium falciparum* (20), often expanding to >50% of total blood T cells (21), and show responses to both HIV (22) and influenza (23). $V\delta 2^{(+)}$ T cells also kill a spectrum of malignant cells that includes leukemias and lymphomas and solid tumors such as renal cell, breast, prostate, and colorectal carcinomas (24). Indeed, meta-analysis of gene expression signatures from ~18,000 human tumors across 39 malignancies indicated a tumor-associated $\gamma\delta$ T-cell profile as the best predictor of patient survival (25). Thus, there appears enormous potential to harness these antipathogen and antitumor effector functions for clinical applications.

Despite this therapeutic promise, results from phase I/II clinical trials that have activated $V\delta 2^{(+)}$ T cells with aminobisphosphonates are mixed. Although objective clinical outcomes were observed in some patients with relapsed/refractory low-grade non-Hodgkin’s lymphoma, multiple myeloma, metastatic hormone-refractory prostate cancer, or advanced metastatic breast cancer (26–28), numerous patients failed to demonstrate effective $V\delta 2^{(+)}$ T-cell responses. Clearly, understanding this person-to-person heterogeneity in $V\delta 2^{(+)}$ T-cell responsiveness, correlated with subsequent clinical outcome, is

Significance

A type of human white blood cell, known as the $V\delta 2^{(+)}$ T cell, has shown promise in immunotherapies against a range of tumors. However, in recent clinical trials patient responses and clinical outcomes have been variable and unpredictable. To address this, we here reveal a significant variability in $V\delta 2^{(+)}$ T-cell functional potential between individuals in the general population, which develops shortly after birth, is stable over time, and is manifested by differential mechanistic capacities to kill tumor targets. These results support personalized clinical approaches to identify patients with “ $V\delta 2$ profiles” that are compatible with killing of their particular tumor and suggest that tailored $V\delta 2$ -profile-specific activation protocols may maximize the chances of future treatment success.

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¹N.S. and C.J.H. contributed equally to this work.

²To whom correspondence should be addressed. Email: d.pennington@qmul.ac.uk.

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critical not only for optimization of $V\delta^{2+}$ T-cell-related therapies, but also for predicting disease progression where $V\delta^{2+}$ T-cell responses are involved.

In this study, we reveal functional $V\delta^{2+}$ T-cell heterogeneity between individuals in a large cohort of healthy volunteers. The effector potentials of these “ $V\delta$ profiles” are characterized by two dominant but qualitatively contrasting phenotypes. At one extreme, $V\delta^{2+}$ T cells display high proliferative capacity, express several cytokine and chemokine receptors, and demonstrate unusual granzyme K-mediated target cell killing. At the other extreme, $V\delta^{2+}$ T cells have lower expansion potential but possess a dominantly cytotoxic nature characterized by granzyme B-mediated cytotoxicity. This interindividual $V\delta^{2+}$ T-cell heterogeneity develops after birth although acquisition of a particular $V\delta$ profile does not correlate with gender, age, country of birth, or chronic $V\delta^{2+}$ T-cell stimulation in vivo. Moreover, these $V\delta$ profiles were stable in individuals over the 3-y study period, suggesting an active homeostatic maintenance. Importantly, an individual's $V\delta$ profile predicts functional potential that we demonstrate by differential killing of various tumor cell lines. Thus, these data highlight a phenotypic and functional heterogeneity in the human $V\delta^{2+}$ T-cell pool that has profound clinical implications such that individuals with different $V\delta$ profiles would be predicted to respond differently to $V\delta^{2+}$ T-cell-targeted immunotherapies or in response to infections.

Results

Significant Functional Heterogeneity in $V\delta^{2+}$ T-Cell Subsets Between Healthy Individuals. We had regularly observed phenotypic heterogeneity when using the commonly used markers CD27 and CD45RA to assess human $V\delta^{2+}$ T cells from small numbers of healthy volunteers (Fig. 1A). As this compromised our interpretation of $V\delta^{2+}$ T-cell involvement in disease, the nature and extent of this heterogeneity was investigated in a much larger cohort of healthy individuals ($n = 63$). In our hands, CD45RA staining of $V\delta^{2+}$ T cells (but not other T-cell subsets) does not give distinct demarcation of positive and negative subsets (Fig. 1A). Thus, we instead assessed $V\delta^{2+}$ T cells using CD27, CD28, and CD16 that consistently identified four distinct $V\delta^{2+}$ populations: $\gamma\delta^{(28+)}$ cells [$CD28^{(+)}CD27^{(+)}CD16^{(-)}$], $\gamma\delta^{(28-)}$ cells [$CD28^{(-)}CD27^{(+)}CD16^{(-)}$], $\gamma\delta^{(16-)}$ cells [$CD28^{(-)}CD27^{(-)}CD16^{(-)}$],

and $\gamma\delta^{(16+)}$ cells [$CD28^{(-)}CD27^{(-)}CD16^{(+)}$] (Fig. 1B). The $\gamma\delta^{(28+)}$ population was most prevalent across the cohort (Fig. 1C) comprising on average $\sim 54\%$ of total $V\delta^{2+}$ T cells, followed by the $\gamma\delta^{(28-)}$ subset ($\sim 22\%$), $\gamma\delta^{(16+)}$ subset ($\sim 11\%$), and $\gamma\delta^{(16-)}$ subset ($\sim 8\%$). However, dominance of a particular subset did not correlate with age, gender, or developmental index of country of birth (Fig. S1). The four $V\delta^{2+}$ T-cell subsets were functionally assessed after stimulation with bisphosphonate (zoledronate) and IL-2. Proliferative potential segregated with CD27 expression, as the $\gamma\delta^{(28+)}$ and $\gamma\delta^{(28-)}$ subsets, but less so the $\gamma\delta^{(16-)}$ and $\gamma\delta^{(16+)}$ subsets, divided multiply after activation (Fig. 1D). Consistent with this, $\gamma\delta^{(16+)}$ cells expressed the highest level of CD57 that has been reported to correlate with replicative senescence (29) although other markers of exhaustion, such as PD-1, were similar across all populations (Fig. S2). All subsets had potential to secrete both IFN γ and TNF α (Fig. S3). However, cytolytic capacity, assessed by intracellular staining for perforin and granzyme B (Fig. 1E), and surface staining of CD56 (Fig. S4), was greater in the $\gamma\delta^{(16-)}$ and $\gamma\delta^{(16+)}$ subsets. Finally, the $\gamma\delta^{(28+)}$ subset demonstrated a reduced capacity to degranulate, judged by surface CD107a expression after 8 h of activation (Fig. 1F). Thus, CD27, CD28, and CD16 unambiguously identify four subsets of $V\delta^{2+}$ T cells in normal healthy individuals. Importantly, as these subsets have distinct functional capacities, the overall effector potential of the $V\delta^{2+}$ T-cell compartment will reflect the relative contributions of these subsets in any given individual.

Individuals Possess Stable $V\delta$ Profiles. The 63 healthy individuals could be stratified into six $V\delta$ profiles defined by relative distribution of the $\gamma\delta^{(28+)}$, $\gamma\delta^{(28-)}$, $\gamma\delta^{(16-)}$, and $\gamma\delta^{(16+)}$ subsets (Fig. 2A), which also did not correlate with gender, with developmental index of country of birth, or, as yet, with a limited assessment of ethnicity (Fig. S5). Profile #2 (50–70% $\gamma\delta^{(28+)}$ cells with 10–35% $\gamma\delta^{(28-)}$ cells) was observed most frequently ($n = 28$), and only two profiles featured a single dominant subset; $\gamma\delta^{(28+)}$ cells were dominant in profile #1 whereas $\gamma\delta^{(16+)}$ cells dominated profile #6 (Fig. 2A). This is consistent with $\gamma\delta^{(28+)}$ and $\gamma\delta^{(16+)}$ cells being at opposite ends of a differentiation pathway, with $\gamma\delta^{(28-)}$ and $\gamma\delta^{(16-)}$ cells in-between. However, the prevalence of any particular profile (e.g., profile #6) did not increase with age (Fig. 2B), suggesting that regular and cumulative exposure to phosphoantigen-producing

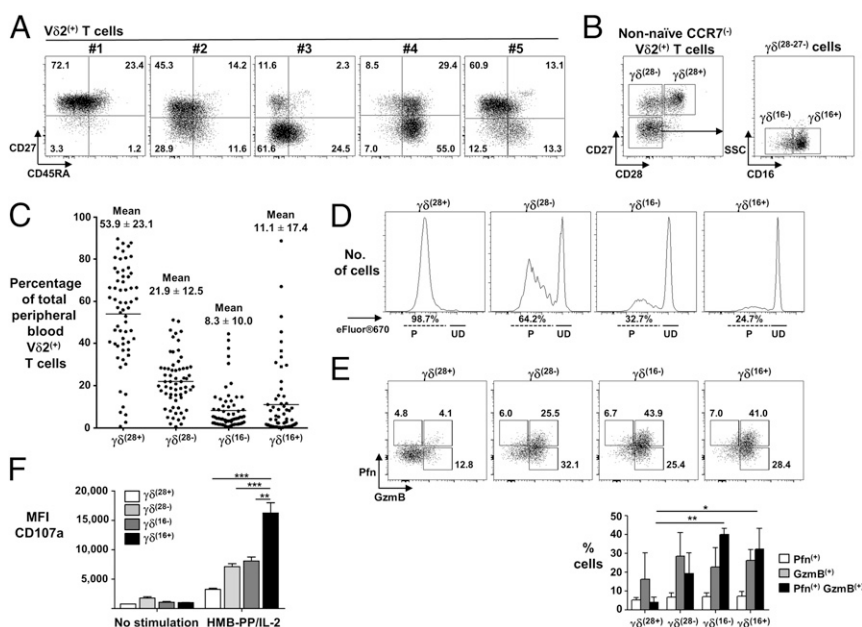


Fig. 1. Interindividual $V\delta^{2+}$ T-cell phenotypic variation. (A) CD27/CD45RA plots of blood $V\delta^{2+}$ cells from five healthy individuals. Quadrant percentages are indicated. (B) Nonnaive CCR7⁺ $V\delta^{2+}$ cells comprise four subsets: $\gamma\delta^{(28+)}$ [$CD28^{(+)}CD27^{(+)}CD16^{(-)}$], $\gamma\delta^{(28-)}$ [$CD28^{(-)}CD27^{(+)}CD16^{(-)}$], $\gamma\delta^{(16-)}$ [$CD28^{(-)}CD27^{(-)}CD16^{(-)}$], and $\gamma\delta^{(16+)}$ [$CD28^{(-)}CD27^{(-)}CD16^{(+)}$]. (C) $V\delta^{2+}$ subset distribution for 63 healthy individuals (mean $\pm$ SD). (D) Representative plots of eFluor670 dilution in sorted $V\delta^{2+}$ subsets after culture with $V\delta$ -depleted mononuclear accessory cells (1:1) 7 d after stimulation with zoledronate (1 μ M) + IL-2 (100 U/mL). P, proliferating cells; UD, undivided cells. (E) Intracellular Perforin and granzyme B in $V\delta^{2+}$ subsets after 4 h stimulation of PBMC ex vivo with phorbol 12-myristate 13-acetate (PMA)/ionomycin. Summary chart for percentage of $V\delta^{2+}$ subset producing. Pfn⁺, perforin; GzmB⁺, granzyme B; both, Pfn⁺GzmB⁺ ($n = 3$). (F) Mean fluorescence intensity (MFI) of CD107a on $V\delta^{2+}$ subsets 8 h after activation of PBMCs with HMB-PP (20 nM) + IL-2 (100 U/mL). Multiple comparison testing using one-way ANOVA with Tukey's post hoc test used in E and F. * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

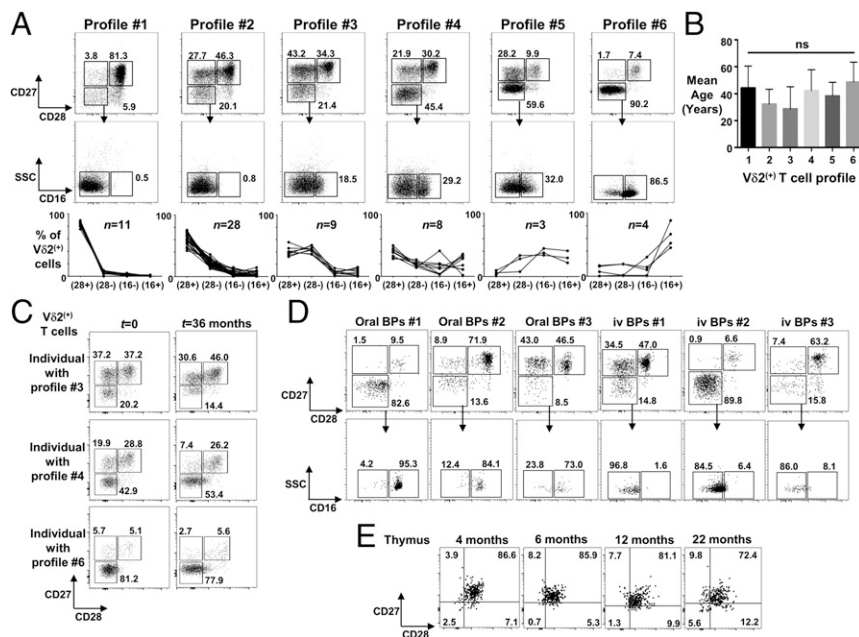


Fig. 2. Healthy individuals have stable Vδ2 profiles. (A, Top rows) Representative individuals possess distinct Vδ2 profiles. (Bottom charts) Individuals ($n = 63$) were assigned to a Vδ2 profile. (B) Chart of mean age (years) for Vδ2 profiles (#1 to #6). (C) Longitudinal analysis of Vδ2 profiles #3, #4, and #6 at $t = 0$ mo and 36 mo. (D) Plots of Vδ2 profiles from osteoporotic patients on weekly oral (oral BPs; $n = 3$) or annual (>3 y) i.v. (iv BPs; $n = 3$) bisphosphonates. (E) Plots of Vδ2(+) T cells from four infant thymuses (4–22 mo old). Percentage of gated cells is indicated.

pathogens is not sufficient to drive differentiation of Vδ2(+) T cells from one subset phenotype to another. This stable nature of an individual's Vδ2 profile over time was supported by a longitudinal analysis of Vδ2(+) T cells from three individuals over 36 mo, in which each person's Vδ2 profile remained largely constant (Fig. 2C). Moreover, analysis of blood from osteoporotic patients on long-term oral or i.v. bisphosphonates (essentially the chronic activation of Vδ2(+) T cells in vivo) did not reveal a selective accumulation of any Vδ2(+) T-cell subset or profile (Fig. 2D). In vitro monitoring over 12 d of Vδ2(+) T cells after phosphoantigen/IL-2 activation of peripheral blood mononuclear cells (PBMCs) from healthy individuals demonstrated significant recovery of the initial Vδ2 profiles (Fig. S6). And even short-term stimulation of sorted Vδ2(+) T-cell subsets by phosphoantigen/IL-2 for 3 d, in the presence of sorted CD14(+) autologous accessory cells, demonstrated little change to the initial Vδ2(+) T-cell phenotypes (Fig. S7). Thus, the Vδ2(+) T-cell compartments of normal healthy adult individuals can be stratified into distinct Vδ2 profiles that are stable over time and that do not essentially change in response to acute or chronic Vδ2(+) T-cell activation. To investigate whether these Vδ2 profiles are "set" during early Vδ2(+) T-cell development, Vδ2 profiles were assessed from 10 neonatal (4–22 mo) thymuses obtained from cardiac surgeries. Only the CD28(+)CD27(+)CD16(−) phenotype of profile #1 was evident in all samples (Fig. 2E). Such initial uniformity of Vδ2(+) T-cell phenotype suggests that the range of distinct Vδ2 profiles is established postthymically, presumably in response to early interactions with the neonatal and/or infant environment.

Vδ2 Profiles Are Polarized Toward Two Dominant Phenotypes. Each Vδ2 profile consists of different proportions of $\gamma\delta^{(28+)}$, $\gamma\delta^{(28-)}$, $\gamma\delta^{(16-)}$, and $\gamma\delta^{(16+)}$ cells, the combination of which will differentially dictate overall Vδ2(+) T-cell responses. To explore these functional potentials further, we sorted $\gamma\delta^{(28+)}$, $\gamma\delta^{(28-)}$, $\gamma\delta^{(16-)}$, and $\gamma\delta^{(16+)}$ cells to high purity from three different individuals to perform microarray expression analysis (using Illumina's HumanHT-12 v4 beadchip array). After multiple comparisons for genes expressed at least twofold higher or lower between any two subsets the $\gamma\delta^{(28+)}$ and $\gamma\delta^{(16+)}$ populations appeared most different (469 genes), consistent with these subsets having distinct functional potentials (Fig. 3A). By contrast, only 78 genes differed between $\gamma\delta^{(28-)}$ and $\gamma\delta^{(16-)}$ cells.

We also reanalyzed our flow cytometry data using Gemstone software that assesses multiple flow parameters from multiple flow cytometry data files simultaneously (www.vsh.com/products/gemstone/). This is used to order or group cells by phenotypic similarity. Using assumptions that CD27, CD28, and CD16 expression can be high or low on any cell, the software analyzed all collected events from all 63 individuals to generate a summary representation of common cell phenotypes from all samples (Fig. 3B). This analysis supported the idea of dominant $\gamma\delta^{(28+)}$ and $\gamma\delta^{(16-)}$ subsets and again suggested less well-defined distinctions between $\gamma\delta^{(28-)}$ and $\gamma\delta^{(16+)}$ cells with several additional intermediate phenotypes. Taken together with our data indicating that Vδ2 profiles are stable over time, it suggests that the Vδ2(+) T-cell compartments of healthy adult individuals are polarized, to different extents, toward one of two major effector potentials, dominated, respectively, by either the $\gamma\delta^{(28+)}$ or $\gamma\delta^{(16+)}$ subset.

Demarcation of Dominant Vδ2(+) T-Cell Phenotypes by CCR6 and CX3CR1.

To explore Vδ2 profile effector potentials further, a table of immune-related genes that differed in expression by $\geq$ twofold between $\gamma\delta^{(28+)}$ and $\gamma\delta^{(16+)}$ cells was compiled (Fig. 3C). This highlighted differential expression in $\gamma\delta^{(16+)}$ cells of genes related to cytotoxic function, including perforin, granzyme B, granzyme H, granzyme I, and eight members of the killer cell Ig-like receptor (KIR) family. By contrast, $\gamma\delta^{(28+)}$ cells expressed higher levels of granzyme K (30) and cytokine and chemokine receptors such as IL-7R α , IL-18R α , IL-23R, CCR2, CCR7, and CCR6 (see below).

Of particular interest was differential expression of CCR6 in $\gamma\delta^{(28+)}$ cells (~ 4.4 -fold higher) and of CX3CR1 (fractalkine receptor) in $\gamma\delta^{(16+)}$ cells (~ 10 -fold higher), which was validated in 12 individuals (Fig. 4A). CX3CR1 was expressed at high levels by all $\gamma\delta^{(16+)}$ cells, but only by a fraction of the $\gamma\delta^{(28+)}$ subset and at a lower per-cell level. Conversely, CCR6 was not detected on $\gamma\delta^{(16+)}$ cells, but identified a distinct population of CCR6(+)CX3CR1(−) $\gamma\delta^{(28+)}$ cells. An identifiable subset of CCR6(+)CX3CR1(+) double-positive cells was never truly evident, suggesting a mutually exclusive relationship that may have functional implications. Consistent with this, the intensity of CX3CR1 expression correlated with cytolytic potential as judged by coexpression of granzyme B and perforin (Fig. 4B). By contrast, CCR6(+) $\gamma\delta^{(28+)}$ cells were GzmB(−)Pfn(−) but expressed granzyme K (Fig. 4C). CCR6(+) $\gamma\delta^{(28+)}$ cells also expressed higher surface levels of both CCR2 and CCR5 (Fig. 4D) and the IL-18R α

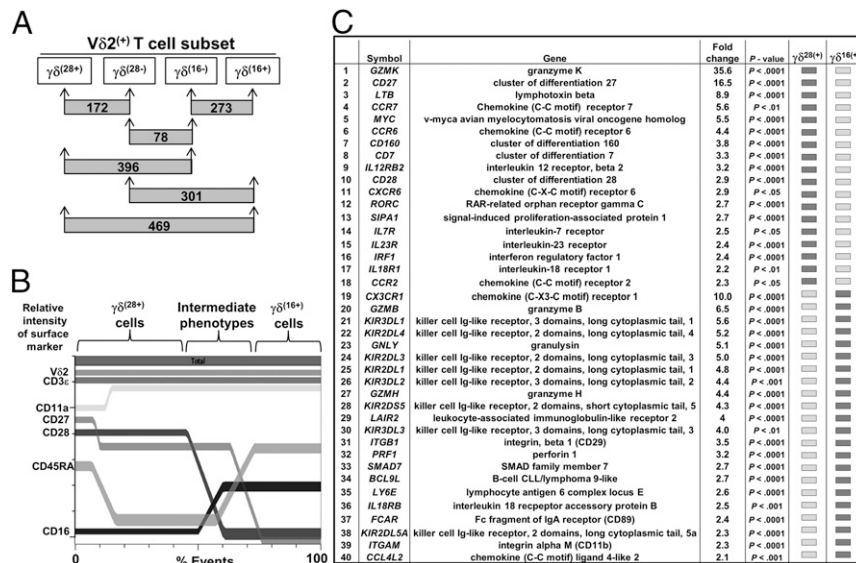


Fig. 3. Vδ2 profiles polarize toward two dominant phenotypes. (A) Number of differentially expressed genes (two- or more fold difference and $P < 0.05$) between Vδ2⁽⁺⁾ subsets. Multiple comparison tests were with false discovery rate Benjamini-Hochberg P -value correction. (B) Average probability state model (Gemstone software) for CD28, CD27, CD16, CD45RA, and CD11a on Vδ2⁽⁺⁾CD3⁽⁺⁾ cells ($n = 63$). Lines show average relative intensity of marker along a cumulative progression axis (x axis). (C) Differentially expressed immune-related genes with two- or more fold expression difference ($P < 0.05$) between γδ⁽²⁸⁺⁾ and γδ⁽¹⁶⁺⁾ subsets. Dark shaded bars indicate in which subset gene expression is highest.

chain (Fig. 4E), compared with CX3CR1⁽⁺⁾ cells. CCR6 and granzyme K were also expressed (in the absence of CX3CR1 and granzyme B) in a sizable proportion of Vδ2⁽⁺⁾ T cells from all our neonatal thymus samples (Fig. 4F), supporting the idea that freshly generated Vδ2⁽⁺⁾ T cells initially adopt a profile #1 phenotype.

CCR6, along with RORγt and IL-23R that are also differentially up-regulated in γδ⁽²⁸⁺⁾ cells (Fig. 3C), are associated with an IL-17A-secreting phenotype (31, 32). Unlike in mice, human γδ cells with IL-17A-secreting potential are not abundant in adult peripheral blood (12). However, they are present in cord blood (33) and in inflammatory lesions of immunopathologies such as meningitis and psoriasis (12, 34); thus, their origin is of

particular importance. As expected, very few IL-17A-secreting Vδ2⁽⁺⁾ T cells were observed in peripheral blood of any individual tested. Nonetheless, when these cells were observed they expressed CCR6 (Fig. 4G) and resided in the γδ⁽²⁸⁺⁾ subset (Fig. S8). Moreover, CCR6⁽⁺⁾ γδ⁽²⁸⁺⁾ cells expressed higher levels of CD161, the up-regulation of which was recently demonstrated on IL-17A-secreting Vδ2⁽⁺⁾ T cells in vitro (12) (Fig. 4H).

Distinct Vδ2 Profiles Display Distinct Cytotoxic Effector Potentials. The data above predict that healthy adult individuals with distinct Vδ2 profiles will display qualitatively different effector

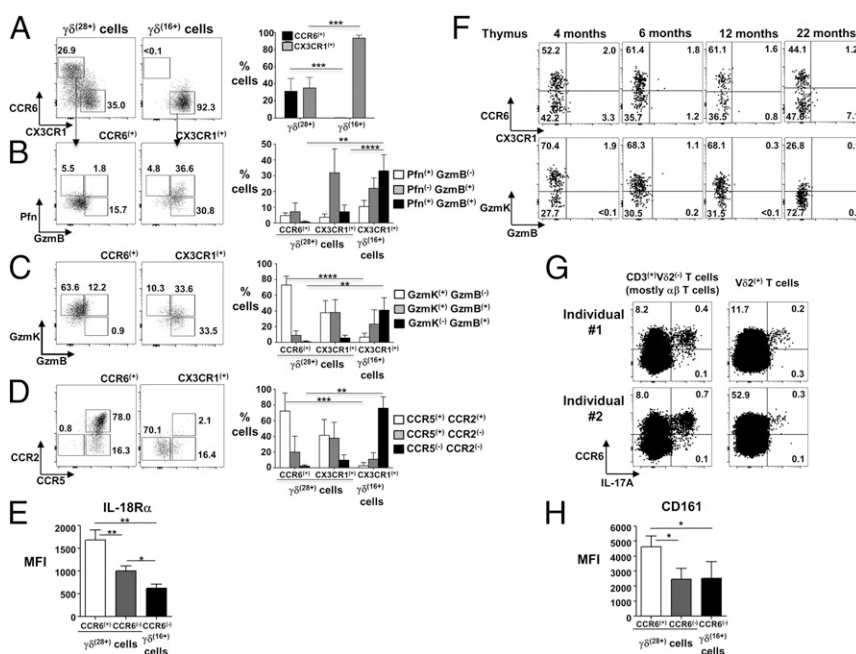


Fig. 4. Demarcation of Vδ2⁽⁺⁾ T-cell phenotypes by CCR6 and CX3CR1. (A) Representative plots (Left) and summary charts (Right) showing (A) CCR6 and CX3CR1 on γδ⁽²⁸⁺⁾ and γδ⁽¹⁶⁺⁾ T cells in healthy individuals ($n = 12$), gating CCR6⁽⁺⁾ γδ⁽²⁸⁺⁾ cells, CX3CR1⁽⁺⁾ γδ⁽²⁸⁺⁾ cells, and CX3CR1⁽⁺⁾ γδ⁽¹⁶⁺⁾ cells. Subsets from A, further stained for intracellular perforin and granzyme B (B), granzyme K and granzyme B (C), or surface CCR2 and CCR5 (D), following 4 h PMA/ionomycin stimulation ($n = 6-9$). (E) MFI for IL-18α for subsets described in A ($n = 5$). (F) Plots for four infant thymuses (4-22 mo old) for CCR6, CX3CR1, and intracellular granzyme K and granzyme B. (G) CCR6 and intracellular IL-17A expression in Vδ2⁽⁺⁾ and Vδ2⁽⁻⁾ (predominantly αβ T-cell) subsets from blood ($n = 2$). (H) MFI for CD161 for Vδ2⁽⁺⁾ subsets described in A ($n = 6$). Mean values with error bars (SD). Multiple comparisons using one-way ANOVA with Tukey's post hoc test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

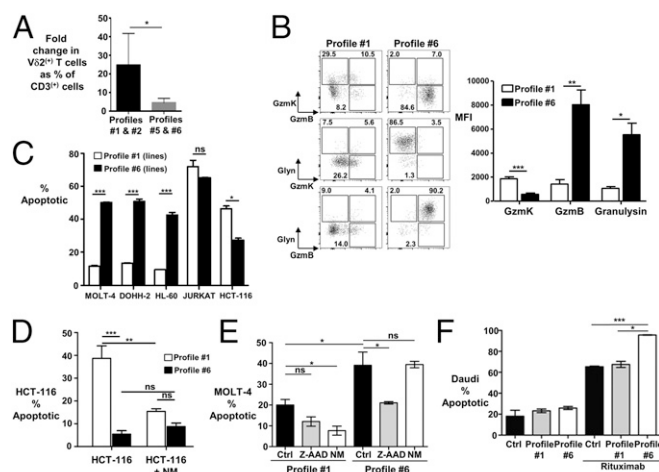


Fig. 5. Distinct V δ 2 profiles display distinct cytotoxic effector potentials. (A) Fold change in V δ 2⁽⁺⁾ cells as percentage of CD3⁽⁺⁾ cells 7 d after stimulation of PBMCs with zoledronate (1 μ M) and IL-2 (100 U/mL), comparing V δ 2 profiles #1 and #2 (n = 10) with profiles #5 and #6 (n = 4). (B) Representative plots and summary MFIs (n = 6) for intracellular granzyme B, granzyme K, and granulysin from sorted V δ 2⁽⁺⁾ cells from V δ 2 profiles #1 or #6 after 4 h stimulation with PMA/ionomycin. (C) Chart showing percentage of apoptosis of tumor lines after 4 h coculture with sorted V δ 2⁽⁺⁾ T cells (V δ 2 profiles #1 or #6) from PBMC cultures stimulated for 12 d with 1 nM HMB-PP and 100 U/mL IL-2. Effector:target ratios were 5:1. Error bars are SD. (D–F) Charts showing percentage apoptosis of PI-9⁽⁺⁾ HCT-116 (D), MOLT-4 (E), or DAUDI (F) cells after 4 h coculture with sorted and activated (1 nM HMB-PP and 100 U/mL IL-2) V δ 2⁽⁺⁾ T cells from V δ 2 profiles #1 or #6 (n = 3) ex vivo. Pretreatment (16 h) of V δ 2⁽⁺⁾ cells with 100 μ M NM was used to inhibit granzyme K (D and E). Granzyme B was inhibited by 100 μ M Z-AAD (E). Rituximab (anti-CD20 mAb) was included to induce antibody-dependent cellular cytotoxicity (F). Effector:target ratios were 5:1. Error bars are SD. Differences compared with two-tailed, unpaired Student's t tests. * P < 0.05, ** P < 0.01, and *** P < 0.001.

responses to phosphoantigen challenge. To explore this, PBMC were isolated from individuals with either V δ 2 profile #1 or #2, in which $\gamma\delta$ ⁽²⁸⁺⁾ cells dominate, or V δ 2 profile #5 or #6, in which $\gamma\delta$ ⁽¹⁶⁺⁾ cells dominate. As expected, total V δ 2⁽⁺⁾ T cells from profiles #1 and 2 expanded on average >10-fold more than those from profiles #5 and 6 in response to HMB-PP/IL-2 (Fig. 5A). Next, we assessed cytotoxic potential by first confirming, in five further subjects, that V δ 2⁽⁺⁾ T cells from profile #6 individuals express high levels of granzyme B and granulysin, whereas V δ 2⁽⁺⁾ T cells from profile #1 individuals instead express granzyme K (Fig. 5B). Consistent with this contrast in cytotoxic potential, HMB-PP/IL-2-expanded V δ 2⁽⁺⁾ T-cell lines from profile #6 subjects efficiently lysed MOLT-4 cells (acute lymphoblastic leukemia), DOHH-2 cells (B-cell lymphoma), and HL-60 cells (acute promyelocytic leukemia) to a much greater extent than expanded V δ 2⁽⁺⁾ T-cell lines from profile #1 individuals (Fig. 5C). Nonetheless, V δ 2⁽⁺⁾ T-cell lines from both profiles efficiently killed Jurkat cells (acute T-cell leukemia), whereas V δ 2⁽⁺⁾ T-cell lines from profile #1 individuals killed HCT-116 cells (colon carcinoma) more effectively than those from profile #6 (Fig. 5C). HCT-116 cells are reported to express proteinase inhibitor 9 (PI-9), a potent inhibitor of granzyme B (but not granzyme K) (35), possibly explaining the reduced cell killing in profile #6 cultures. This differential killing was confirmed using sorted V δ 2⁽⁺⁾ T cells from profile #1 and #6 individuals activated with HMB-PP/IL-2 for 4 h and then cocultured with HCT-116 cells (Fig. 5D). Importantly, the greater killing of HCT-116 cells in profile #1 cultures could be inhibited by preincubation of the V δ 2⁽⁺⁾ T cells with nafomostat mesylate (NM) that inhibits granzyme K (36) (Fig. 5D). NM also reduced killing of MOLT-4 cells by freshly isolated HMB-PP/IL-2-activated V δ 2⁽⁺⁾ T cells from profile #1 individuals, whereas granzyme B inhibitor Z-Ala-Ala-Asp(OMe)-chloromethyl ketone (Z-AAD) (37) had minimal effect (Fig. 5E).

Conversely however, Z-AAD, but not NM, was able to inhibit MOLT-4 cell killing in similar cultures of activated V δ 2⁽⁺⁾ T cells from profile #6 individuals, confirming a granzyme B-mediated mode of cytotoxicity (Fig. 5E). V δ 2⁽⁺⁾ T cells from profile #6 subjects also express abundant CD16 that can mediate antibody-dependent cellular cytotoxicity. Consistent with this, DAUDI cells (Burkitt's lymphoma) that are killed to some degree by the anti-CD20 antibody rituximab, show almost complete cell lysis in the presence of activated V δ 2⁽⁺⁾ T cells from profile #6 individuals (Fig. 5F). By contrast, V δ 2⁽⁺⁾ T cells from profile #1 subjects in the presence of rituximab, and V δ 2⁽⁺⁾ T cells from either profile in the absence of rituximab, displayed no significant DAUDI cell killing above control (Fig. 5F). In sum, our data demonstrate that individuals with distinct V δ 2 profiles display qualitatively distinct V δ 2⁽⁺⁾ T-cell-mediated cytotoxic effector function that is mediated in large part by differential expression of cytotoxic mediators such as granzyme B and K. Importantly, this suggests that only individuals possessing V δ 2 profiles that are compatible with particular features of their tumor targets are likely to show promising clinical outcomes.

Discussion

Current understanding of V δ 2⁽⁺⁾ T-cell biology does not adequately explain the variability in objective clinical responses in V δ 2⁽⁺⁾ T-cell-based immunotherapies against cancer (26–28). In addressing this, we demonstrate significant interindividual heterogeneity in V δ 2⁽⁺⁾ T-cell phenotype that translates to distinct V δ 2⁽⁺⁾ T-cell effector potentials. Our analysis of 63 healthy individuals revealed several V δ 2 profiles, each defined by differing proportions of V δ 2⁽⁺⁾ T-cell subsets with distinct functional characteristics. Using multiple analyses, our data suggest that these V δ 2 profiles consist of varying contributions from two dominant but contrasting V δ 2⁽⁺⁾ T-cell populations identified as $\gamma\delta$ ⁽²⁸⁺⁾ and $\gamma\delta$ ⁽¹⁶⁺⁾ cells. Moreover, these opposing phenotypes segregate by mutually exclusive expression of chemokine receptors CCR6 and CX3CR1. CX3CR1 correlates with increased granzyme B/perforin-associated cytotoxic capacity but decreased proliferative potential, whereas CCR6 characterizes $\gamma\delta$ ⁽²⁸⁺⁾ cells with increased expression of cytokine receptors (e.g., IL-18R α) and chemokine receptors (e.g., CCR2 and CCR5) and expression of granzyme K. CCR6⁽⁺⁾ $\gamma\delta$ ⁽²⁸⁺⁾ cells also include the few IL-17A-secreting V δ 2⁽⁺⁾ T cells found in peripheral blood.

A key finding is that distinct V δ 2 profiles appear stable over time, as highlighted in three individuals over 36 mo. It had been suggested from studies in vitro that phosphoantigen stimulation of V δ 2⁽⁺⁾ T cells resulted in differentiation from a CD27⁽⁺⁾CD45RA⁽⁺⁾ phenotype to either a CD27⁽⁻⁾CD45RA⁽⁻⁾ or CD27⁽⁻⁾CD45RA⁽⁺⁾ phenotype via a CD27⁽⁺⁾CD45RA⁽⁻⁾ intermediate (15). However, in our cohort, older individuals, who inevitably will have been exposed to more phosphoantigen-producing pathogens than younger individuals, did not favor any particular V δ 2 profile. Moreover, patients on long-term (>36 mo) bisphosphonate therapies, either orally or intravenously, did not present with any specific V δ 2 profile, as would be expected if repeated stimulation drove V δ 2⁽⁺⁾ T-cell differentiation. Together, these findings suggest V δ 2 profiles change little over time despite repeated stimulation. This evokes a model in which V δ 2⁽⁺⁾ T-cell activation expands short-lived effector cells, but preserves proportions of longer-lived “memory” cells to maintain the V δ 2 profile. The underlying mechanism for this requires further investigation but may involve “memory stem cell” activity, akin to that described for memory CD8⁽⁺⁾ $\alpha\beta$ T cells (38), although whether V δ 2⁽⁺⁾ T cells demonstrate true memory is still uncertain (39).

The stable yet heterogeneous nature of V δ 2 profiles between healthy individuals raises the question of how they develop. The neonatal (4–22 mo) thymus data demonstrate a uniformity of phenotype that resembles profile #1, with a predominance of $\gamma\delta$ ⁽²⁸⁺⁾ cells, and expression of CCR6 and granzyme K. This suggests that, presumably early in life (40), an individual's V δ 2⁽⁺⁾ T-cell compartment transitions from a uniform thymic profile to

a particular peripheral profile that constitutes a stable set-point. The drivers of this transition may be genetic. Alternatively, it may result from childhood exposure to specific pathogens, such as *M. tuberculosis* (19), *P. falciparum* (20), or CMV (41). Such infections may cause aggressive, or qualitatively different, Vδ2⁽⁺⁾ T-cell responses that force a change in Vδ2 profile. Such profile-changing responses could also depend on certain Vγ9Vδ2TCR specificities that are restricted to certain individuals. These ideas are currently under investigation.

A corollary to identification of distinct stable Vδ2 profiles across healthy subjects is that these individuals should demonstrate distinct Vδ2⁽⁺⁾ T-cell-mediated effector responses. Indeed, we demonstrate differences in both efficiency and mechanism by which Vδ2⁽⁺⁾ T cells, from profile #1 versus profile #6 individuals, kill a range of tumor lines. Profile #6 Vδ2⁽⁺⁾ T cells show granzyme B-mediated cytotoxicity, but are not efficient at killing tumor lines that express the granzyme B inhibitor PI-9. However, these cells can additionally deploy ADCC to kill tumor targets in the presence of tumor-binding antibodies. By contrast, Vδ2⁽⁺⁾ T cells from profile #1 individuals favor granzyme K-mediated cytotoxicity and appear more effective at killing PI-9⁽⁺⁾ tumor cells than profile #6 Vδ2⁽⁺⁾ T cells. These data thus have important clinical implications for patient stratification in Vδ2⁽⁺⁾ T-cell-based therapies, as there is likely an optimal Vδ2⁽⁺⁾ T-cell response for any given malignancy. Although the Vδ2⁽⁺⁾ T-cell compartment is in theory capable of making a

range of responses, a particular patient may be restricted in the responses that they can actually make. If this effector potential is mismatched to that required to target a particular tumor, treatment is likely to fail, which may have contributed to inconsistent patient responses in recent clinical trials (26–28, 42). By contrast, selecting patients with defined Vδ2⁽⁺⁾ T-cell effector potentials that complement features of a tumor, and tailoring activation protocols to optimize specific responses, may maximize the chances of future treatment success.

Materials and Methods

Informed consent and ethical approval was obtained for blood [London City and East Research Ethics Committee (REC) 13/LO/0548] and thymus samples (London Queen Square REC 14/LO/2132).

The isolation of PBMCs, flow cytometry, proliferation assays, cytotoxicity assays, degranulation assay, Gemstone probability state modeling, microarray analysis, and statistical analyses are described in *SI Materials and Methods*.

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